

Assessing the Efficiency of Transformer Models with Varying Sizes for Text Classification: A Study of Rule-Based Annotation with DistilBERT and Other Transformers

Arafet Sbei ^{*,§}, Khaoula ElBedoui ^{†,‡} and Walid Barhoumi ^{†,‡}

^{*}*ICube CNRS UMR 7357 University of Strasbourg, Strasbourg, France*

[†]*University of Tunis El Manar, Higher Institute of Computer Science,
Research Team on Intelligent Systems in Imaging and Artificial Vision (SIIVA),
LR16ES06 Computer Science Research Laboratory
Modeling and Processing of Information and Knowledge (LIMTIC),
2 Bayrouni Street, 2080 Ariana, Tunisia*

[‡]*University of Carthage, National School of Engineers of Carthage,
45 Entrepreneurs Street, 2035 Tunis-Carthage, Tunisia
ssbeiiarafett@gmail.com*

Received 31 July 2024

Revised 21 October 2024

Accepted 1 November 2024

Published 23 December 2024

This study presents a comparative analysis of transformer models for text classification, utilizing a hybrid approach that integrates rule-based regular expressions with fine-tuned neural network models. Initially, regular expressions are employed to annotate sentences in a cost-effective manner, providing an efficient alternative to manual labeling. The annotated dataset, comprising around 33,000 instances across three classes (Reminder, Scheduled Activity, General) and also restructured into two classes (Reminder and General) by merging “Scheduled Activity” with “Reminder”, is then used to fine-tune various transformer models, including DistilBERT, BERT, RoBERTa, ALBERT, Electra, Ernie 2.0, XLNet, and GPT-2. Our methodology involves freezing all layers except the final one during fine-tuning, allowing the models to learn nuanced linguistic patterns while mitigating overfitting. Results reveal that DistilBERT, despite its smaller size (66 million parameters), outperforms larger models such as BERT and GPT-2 in terms of accuracy, precision, recall, and F1-score. Likewise, we demonstrated that the proposed method can work better than generative AI large language models for both zero-shot and one-shot learning, namely GPT-3.5, GPT-4, GPT-4o and LLaMA-3 70B. This efficiency is attributed to the distillation process that retains essential features while reducing computational demands. Notably, DistilBERT achieved an overall accuracy of 0.86, significantly surpassing BERT’s 0.55, GPT-2’s 0.36, XLNet’s 0.51, Ernie 2.0’s 0.72, Electra’s 0.74, ALBERT’s 0.72, and RoBERTa’s 0.71. The study highlights the importance of model size and architecture in achieving optimal performance, especially in resource-constrained

[§]Corresponding author.

This is an Open Access article published by World Scientific Publishing Company. It is distributed under the terms of the [Creative Commons Attribution 4.0 \(CC BY\) License](https://creativecommons.org/licenses/by/4.0/), which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

scenarios. This investigation underscores the efficacy of combining rule-based methods with advanced transformer models for text annotation, demonstrating that a balanced approach leveraging both handcrafted rules and learned representations can generalize better than relying solely on one technique. The proposed hybrid method offers a robust and adaptable solution for sentence annotation pipelines, enhancing performance in diverse natural language processing applications with limited labeled data. Code is available at <https://github.com/arafet/Text-annotation-using-rule-based-method-and-Transformers>.

Keywords: Data annotation; rule-based methods; transformers; hybrid labeling; text classification.

1. Introduction

Sentence classification is a fundamental task within the realm of Natural Language Processing (NLP). It aims to improve the ability of machines to understand and process human language, making it an essential element in the advancement of NLP technologies. Indeed, sentence classification involves categorizing sentences into predefined classes based on their content and purpose. In fact, by effectively classifying sentences, it is possible to extract meaningful insights from large datasets, enabling more efficient data analysis and decision-making processes. This process is fundamental for various applications, including sentiment analysis,¹ spam detection,² and topic categorization.³ For instance, within the framework of sentiment analysis, sentence classification allows to determine the sentiment of customer reviews, which can guide businesses in improving their products and services. Likewise, in spam detection, it helps identify and filter out unwanted messages, ensuring a cleaner communication environment. Moreover, sentence classification aids in topic categorization by organizing text into thematic groups, facilitating better information retrieval and content management. Overall, traditional sentence classification methods often rely on task-tailored annotated datasets,⁴ where each sentence is meticulously labeled with its corresponding class or category. However, in the real world, obtaining such annotated datasets can be a daunting and resource-intensive challenge. Annotators must understand the context and accurately label the text, which can be challenging when dealing with idiomatic expressions, sarcasm, or complex sentence structures. Furthermore, any mistakes or inaccuracies in the annotation process can significantly impact the performance of the resulting models.⁵ To this end, several techniques have been developed to automate the data annotation process, but these often fall short.⁶ For instance, unsupervised learning methods can be expensive and may not provide the desired outcomes due to the lack of defined labels.⁷ On the other hand, supervised learning methods require a large amount of labeled data, which may not always be available.⁶ In light of these challenges, this paper is built upon our previous work presented at the Asian Conference on Intelligent Information and Database Systems.⁸ In fact, it investigates a novel framework using regular expressions combined with transformer models to perform an efficient text annotation. Leveraging the representations learned by transformer models can potentially overcome limitations of manually defined rules, while retaining

interpretability. Specifically, our objective is to categorize sentences into three distinct classes: ‘Reminder’, ‘Scheduled Activity’, and ‘General’. The ‘Reminder’ category encompasses tasks that need to be accomplished. These tasks, however, do not necessarily have a specific time frame associated with them. On the other hand, ‘Scheduled Activity’ refers to meetings, events, appointments, and calls that are time-bound and occur at a predetermined date and time. The ‘General’ category includes sentences that do not fit into the specific categories of reminders or scheduled activities. The main contributions of this work are as follows:

- We propose a hybrid method that leverages the capacity of rule-based annotation on dealing with the most common cases and transformers on re-annotating complex sentences.
- We generated a labeled dataset derived from the Enron Emails dataset,⁹ providing a foundational resource for future research focused on classifying text into categories such as scheduled activities, reminders, and general content. This would be beneficial in different NLP applications that needs automatic detection of important tasks.
- Through experiments conducted on two available datasets, we demonstrate that our proposed method achieves state-of-the-art performance while being less complex.

The rest of the paper is organized as follows. In Sec. 2, existing annotation techniques presented in the literature are reviewed. The proposed method is detailed in Sec. 3. The performance of the proposed data labeling method is investigated in Sec. 4. In Sec. 5, the performance of DistilBERT is compared with other classification models and generative AI models. Finally, conclusions are drawn, and ideas for future work are presented in Sec. 6.

2. Related Work

Text classification involves assigning text documents to one or more predefined categories or classes. Despite extensive researches, building text classifiers remains challenging due to the complexity of language and the effort required to obtain quality training data. In fact, this has led to several attempts on labeling textual data for various downstream tasks. The most common annotation approach is the manual labeling which requires human annotators to read documents and assign labels based on guidelines.¹⁰ For instance, various web based annotators have been proposed on the literature notably, FLAT¹¹ which provides linguistic and semantic-based annotation using Folia format to annotate the biomedical document. TeamTat¹² introduces an open-source web-based document annotation tool capable of annotating both plain text inputs and document inputs (in BioCXML or XML format). The resulting output document is presented in the form of an inline annotated BioCXML document. For more text manual annotation tools, readers can refer to

Ref. 6. However, this process is time-consuming, expensive, and prone to subjectivity and errors. Given the nuances of language, even expert annotators may disagree or make inconsistent judgments, reducing dataset quality. Furthermore, manual methods are difficult to scale to the volume of data needed for modern Machine Learning (ML). Thus, semi-supervised techniques, essentially based on Active Learning (AL), have been developed.¹³ In Ref. 14, the Tagtog tool provides annotation at both entity and document levels based on AL. Schroder *et al.*¹⁵ investigated AL techniques based on uncertainty strategies using transformer-based models. They also presented empirical results for text annotation. In Ref. 16, semi-automatic multi-labeled annotation is achieved through the process of selecting the base corpus, automatic annotation through word matching and weight calculation, and manual correction in case of multiple emotional tendencies are found. A semi-supervised method which involves leveraging various machine learning algorithms, including Support Vector Machine, Decision Tree, K-Nearest Neighbors, and Naive Bayes, in conjunction with Word2Vec and TF-IDF text extraction has been proposed in Ref. 17. However, text analysis based on Word2Vec and TF-IDF relies heavily on the used vocabulary. Despite their efficiency on reaching state-of-the-art performances when labeling texts, semi-supervised methods depend strongly on the pre-labeled data as well as the used classifier. Thus, a transformer-based model may fail to properly annotate data with few examples, which requires an extensive human intervention to properly align the model with the desired labeling. Authors in Ref. 18, integrated human annotators and generative AI models within an Active Learning framework. In order to reduce human intervention, the authors relied on uncertainty level given by the Large Language Model (LLM). Given sufficient training data, supervised classifiers utilizing ML algorithms tend to be more robust and accurate. Models implicitly learn textual features and internal representations. State-of-the-art neural networks like BERT²⁸ produce highly flexible text encodings. However, supervised approaches require massive labeled datasets which can be challenging and expensive to produce. Furthermore, unsupervised learning clusters unlabeled documents based on similarity features, rather than predefined labels.¹⁹ It saves the costs of annotation, but the discovered groupings may not align with desired classifications. In Ref. 20, Rasmuss *et al.* introduced a novel approach known as 'super-unsupervised' learning to address the challenges of measuring complex psychological constructs from vast amounts of text data. This method uniquely blends the strengths of both supervised and unsupervised learning techniques to assess constructs like online political hostility without relying on any labeled data. Lacking supervisory signal, unsupervised models also tend to be less accurate. Weak or ambiguous clusters make separating relevant classes challenging. For highly imbalanced data, however, unsupervised techniques may help identify rare classes. In many cases, rule-based classifiers have relied on feature engineering that include grammatical rules to find and classify properties of text. NegEx,²¹ a popular component of rule-based methods, uses regular expressions for detecting negation of findings. The advantage is interpretability — analysts can understand why decisions were made.

However, rules fail to capture contextual nuances and may overly rely on surface patterns. Frequent updates are needed as language evolves. In fact, hybrid methods that integrate rule-based systems with representation learning combine the benefits of human knowledge and ML based models. For instance, a BERT model trained on existing annotated texts coupled with experts annotations, has been proposed in Ref. 22. The model showed prominent results in comparison with state-of-the-art methods. Nevertheless, producing high-quality annotated corpora remains a key obstacle. Recently, a text annotation method that combines regular expression with DistilBERT has been proposed.⁸ The method showed prominent results when compared to generative AI-based annotation. In recent months, the development and advancements of LLMs such as Claude 2.0 and GPT-4 have garnered significant attention from researchers, particularly for their growing impact on applications like text classification. Leveraging their text generation capabilities, generative AI models have been adapted to function as classifiers. For instance, Zhong *et al.*²³ utilized a prompting approach with a zero-shot strategy to annotate sentiment without training examples. In another study, a prompt template that incorporates segments of source code was employed to enhance the effectiveness of source code classification.²⁴ A novel iterative prompt template engineering approach, discussed in Ref. 25, integrates files by exploiting prompts and assesses the quality of responses provided by the Claude 2.0 LLM. However, relying solely on generative AI models is not always reliable due to their propensity to hallucinate, leading to inaccurate outputs. Moreover, generative AI models have also been coupled with distilled classification models.²⁶ The role of LLMs was to generate labeled data, which was subsequently used to train classification models. Nevertheless, such an operation strongly depends on the quality of the generated text, which cannot guarantee a valuable similarity to real-world data.

3. Methods

In this section, we describe the followed process for the proposed annotation method (Fig. 1). In fact, we proceed by describing the data preparation process. Then, we detail the rule-based method for labeling text instances. Finally, the Transformer-based annotation is described.

3.1. Data preparation

The Enron Email Dataset contains over 500,000 emails, however some entries have missing data. To prepare a clean dataset, any rows with empty columns were first removed. From this filtered set, the first 60,000 emails were extracted for further processing. These selected emails were tokenized into sentences using the Natural Language Toolkit (NLTK) sentence segmentation module.²⁷ This module automatically splits text into sentences based on punctuation, context, and other heuristics. In total, the 60,000 emails yielded 894,707 sentences after segmentation.

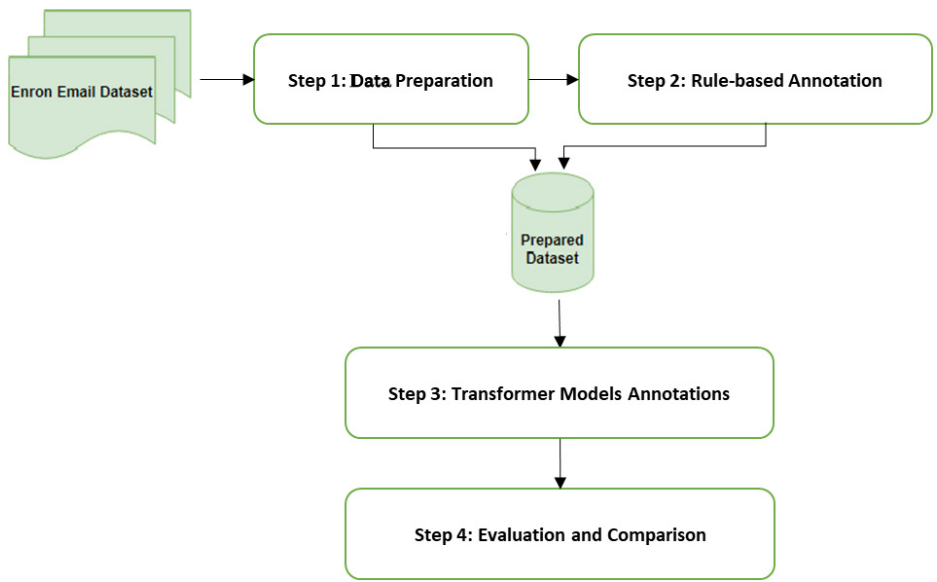


Fig. 1. Outline of the proposed method.

Next, each sentence was preprocessed by first replacing newline characters, then stripping any excess whitespace, and finally lowercasing all text. These cleaning and normalization steps help remove formatting inconsistencies and allow the model to focus on linguistic patterns rather than capitalization or punctuation nuances. In the end, this process produced a cleaned corpus of 894,707 sentences derived from 60,000 Enron emails ready for further analysis and modeling. Then, each input phrase is tokenized into individual words. Finally, each word in the tokenized phrase is tagged with its part of speech using NLTK’s Part of Speech (POS) tagger.²⁹

3.2. Rule-based annotation

The rule-based method for labeling data related to scheduled activity, reminder and general classes represents the first part of our annotation approach. It leverages regular expressions to identify phrases that indicate the three studied classes.

3.2.1. Scheduled activity label

The “scheduled activity” class uses two regular expression patterns to match sequences of tags and keywords. On the one hand, the first pattern is given by:

$(?:\backslash w+\backslash s+)+(?:VBZ\backslash s+VBG|MD\backslash s+VB)\backslash s+(?:\backslash w+\backslash s+)*(?:\text{appointment}|\text{meet}|\text{meeting}|\text{call}|\text{reunion}|\text{event})+\backslash b(?:\backslash w+\backslash s+)*\backslash b$

This matches sequences where a verb in the present tense in the third person (VBZ) singular is followed by a gerund (VBG), or a modal verb (MD), such as “will”, is followed by a base verb (VB). In fact, both cases should be followed by one of the specified keywords (e.g. “appointment”, “meet”). For example, in the phrase “John is setting up a meeting”, “is” is a VBZ and “setting” is a VBG, which represents the first case, while the sentence “I will have a meeting” corresponds to the second. On the other hand, the second pattern is expressed as follows.

$$(\text{?}:\backslash\text{w}+\backslash\text{s}+)\text{+VBP}\backslash\text{s}+(\text{?}:\backslash\text{w}+\backslash\text{s}+)*(\text{?}:\text{appointment}|\text{meet}|\text{meeting}|\text{call}|\text{reunion}|\text{event})+\backslash\text{b}(\text{?}:\backslash\text{w}+\backslash\text{s}+)*\backslash\text{b}$$

This last stands for sequences where a verb in the present tense (VBP) is followed by one of the specified keywords. For example, in the phrase “I have an appointment”, “have” is a VBP, which matches the pattern.

3.2.2. *Reminder label*

In what concerns the “reminder” label, we considered both cases of treating tagged sentences or chunks. In fact, chunking is the process of extracting parts of the text following a pre-defined strategy. In the context of NLTK library, chunks work on top of POS-tagging library. Hence, we propose a set of rules that extract chunks from the text, which we use thereafter solely or in combination with other rules to achieve the desired labeling. To this end, the performed chunking intends to identify two specific types which are verbal sentences and question tags denoted as VB-chunk and QT-chunk, respectively. Table 1 gives a detailed description of each type of chunks. Thus, in order to label data as a reminder, we followed the given rules.

- If the sentence starts with a verb in base form (VB) or a modal verb (MD), it is identified as an imperative sentence. For example, in the sentence “Talk tomorrow”, “Talk” is a VB, which matches the first case, while in the sentence “could you talk”, the second case is verified.
- If the first chunk of the sentence is a VB-chunk, it is identified as an imperative sentence. For example, in the sentence “Please go”, “Please” is a UH and “go” is a VB, which forms a VB-chunk.
- If the sentence is a question and contains the word “please”, and starts with a verb in base form or a modal verb, it is identified as an imperative sentence disguised as a question. For example, in the sentence “Open the doors, please?”, “Open” is a VB and the sentence contains “please”, which matches the rule.
- If the sentence is a question and the last chunk is a QT-chunk, and the sentence starts with a verb in base form or a VB-chunk, it is identified as an imperative sentence disguised as a question. For example, in the sentence “Stop it, will you?”, “Stop” is a VB and “will you?” is QT-chunk, which matches the rule. For all

Table 1. Rule-based chunk extraction.

Chunk grammar rule	Description	Example sentence	Chunk
VB-Phrase: {⟨DT⟩⟨,⟩*⟨VB⟩}	Matches a determiner followed by zero or more commas and a verb in base form.	"The, run quickly."	"The, run"
VB-Phrase: {⟨RB⟩⟨VB⟩}	Matches an adverb followed by a verb in base form.	"Always remember your keys."	"Always remember"
VB-Phrase: {⟨UH⟩⟨,⟩*⟨VB⟩}	Matches an interjection followed by zero or more commas and a verb in base form.	"Well, go ahead."	"Well, go"
VB-Phrase: {⟨UH⟩⟨,⟩⟨VBP⟩}	Matches an interjection followed by a comma and a verb in non-3rd person singular present.	"Oh, am I late?"	"Oh, am"
VB-Phrase: {⟨PRP⟩⟨VB⟩}	Matches a personal pronoun followed by a verb in base form.	"You remember the time."	"You remember"
VB-Phrase: {⟨NN.⟩ + ⟨,⟩*⟨VB⟩}	Matches one or more nouns followed by zero or more commas and a verb in base form.	"Dog, fetch the ball."	"Dog, fetch"
Q-Tag: {⟨,⟩⟨MD⟩⟨RB⟩*⟨PRP⟩⟨.⟩*}	Matches a comma followed by a modal verb, zero or more adverbs, a personal pronoun, and zero or more punctuation marks.	"You're coming to the party, aren't you?"	"aren't you"

sentences that does not match with the above-mentioned labels, they are automatically labeled as “general”. In fact, this method provides a robust way to label data based on linguistic patterns. It can be easily extended or adapted to label other types of data by modifying the regular expressions and keywords. Nevertheless, labeling using only rules would fail with complicated phrases which makes the add of ML-based technique necessary to learn more details from data.

3.3. Transformer models annotation

In this study, we employed a range of transformer models for text classification to investigate the impact of model size and structure on performance. We opted for DistilBERT,³⁰ BERT, RoBERTa,³¹ ALBERT,³² and GPT-2,³³ Ernie 2.0,³⁴ Electra³⁵ and XLNet,³⁶ each representing a different configuration of parameters and architecture complexity, thereby fostering a comprehensive understanding of how these factors influence model efficacy. DistilBERT, with its 66 million parameters, is a distilled version of BERT. It retains 95% of BERT’s performance while being smaller, faster, and more efficient, making it a strong candidate for scenarios where computational resources are limited. BERT, the original transformer model with

110 million parameters, serves as a benchmark due to its robust performance across various NLP tasks. It serves as a baseline for comparing the performance of the other models. RoBERTa, particularly in its large variant with 355 million parameters, exemplifies a model optimized for performance at the cost of increased computational demand. Its inclusion helps us understand the trade-offs between model size and performance improvements. ALBERT, with its base version having only 11 million parameters, incorporates parameter-sharing techniques to reduce model size while attempting to maintain performance. It offers insight into the effectiveness of parameter reduction strategies. GPT2, with 1.5 billion parameters, is primarily designed for generative tasks. To adapt GPT2 for classification, we used its scoring mechanism as a classifier. Ernie 2.0, with 103 million parameters, utilizes continual multi-task learning, integrating external knowledge into its training process, which can improve classification performance, particularly in domain-specific tasks. Electra, with 110 million parameters in its base model, takes a different approach from BERT by training models to distinguish between real and replaced tokens, leading to more sample-efficient training and potentially better performance in certain scenarios. Finally, XLNet, which has 340 million parameters, leverages a permutation-based training approach, enabling it to capture bidirectional context more effectively than traditional autoregressive models. For each model, we employed a rule-based approach to annotate the dataset before training on the labeled data. The dataset was split into 90% for training and 10% for validation, with an equal distribution of instances across classes to ensure balanced evaluation. All layers of the base models, except the classification layer, were frozen, allowing us to fine-tune only the classification component and adapt the pre-trained language understanding capabilities to our specific task. This setup enabled a focused assessment of each model's performance in handling the classification task, considering both the size of the training data and the model architectures.

4. Experimental Results

In the first part of this section, we present the data used for the development and validation of the proposed method, followed by details of the implementation. Then, we assess the performance of the rule-based method in properly identifying the learned classes. The classification results of different classification models is reported thereafter for both three-class and binary classification schemes. Finally, results from GPT generative AI models along with LLaMA-3 70B, prompted as classifiers, are compared to our work for both zero-shot and one-shot learning.

4.1. Data and implementation details

4.1.1. Data

This work utilizes the Enron Email Dataset,⁹ a large public dataset containing over 500,000 real emails from executives of the Enron Corporation. These data were

originally collected by the Federal Energy Regulatory Commission during its investigation of Enron after the company's bankruptcy in 2001. The dataset contains emails from 150 users mostly belonging to Enron's senior management over a period spanning from 1998 to 2002. In total, over 500 MB of data were available including raw emails, attachments, and relevant metadata like sender, recipients, subject line, and date. The emails cover a diverse range of business topics and provide visibility into Enron's operations, culture, and accounting practices that led to its collapse. The Enron Dataset represents a unique opportunity to study internal communications of a large corporation based on natural language over an extended timeframe. For this research, a subset of the Enron data, focusing on executive emails, is utilized to study business language and relationships. This real-world business email data provides an interesting test case for studying social dynamics and developing new NLP techniques. Moreover, we considered a chat dataset which comprises 114 dialogues with a total of 300 sentences extracted from a task-oriented chatbot dialogue dataset.³⁷ The chat dataset is less formal than the emails dataset, making prediction more challenging due to the casual and varied language used in chat. In fact, in Ref. 37, annotators identified the "task" class which represent in our case both "Reminder" and "Scheduled activity" classes. To address this, we reconsidered the annotation of the "Task" class to better align with the specific classes studied in our work. As a result, we ended up with 198 instances for the "General" class, 16 for the "Scheduled Activity" class, and 86 for the "Reminder" class. Furthermore, due to the low number of 'Scheduled Activity' instances, we also considered a binary classification task as labeled in the original dataset, renaming the 'Task' class to 'Reminder'.

4.1.2. Implementation details

For the conducted experiments, the used parameters, are summarized as follows. For all models, based on our empirical experiments, the number of batches is set to 16. The learning rate is set to $2e-5$ and then reduced to zero gradually under Consine-Linear policy. For the DistilBERT, ROBERTA, ALBERT, Ernie 2.0, Electra and XLNet models, we set the number of epochs to 70, while a value of 90 is defined for both GPT2 and BERT models. An early stopping is performed when no improvement is recorded.

4.2. Evaluation metrics

Let True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN) represent the following: TP corresponds to the correctly classified positive cases; TN refers to the correctly classified negative cases; FP signifies the misclassified positive cases; and FN denotes the misclassified negative cases. These components form the basis for evaluating the performance of the proposed method through various metrics. Accuracy is defined as the proportion of true results, including both true positives and true negatives, out of the total number of cases. It is

calculated using (1):

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}. \quad (1)$$

Precision measures the accuracy of positive predictions, which is the proportion of true positive predictions relative to the total number of positive predictions, including false positives. The formula for precision is given by

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \quad (2)$$

Recall evaluates the classifier's ability to identify all positive instances, indicating the proportion of actual positives that were correctly predicted. It is calculated as

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \quad (3)$$

F1-score provides a balanced measure of precision and recall by computing their harmonic mean. This metric reflects both precision and recall in a single value, calculated by

$$\text{F1-score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (4)$$

Each metric offers a unique perspective on the performance and accuracy of the method, contributing to a comprehensive evaluation of its effectiveness.

4.3. Performance of the rule-based method

We evaluated the effectiveness of our rule-based method for data labeling by examining the training process performance using different models. As depicted in Fig. 2, both the training and validation loss curves converge towards zero for all the studied models, with DistilBERT showing the best performance, indicating that the DistilBERT classifier has successfully learned to distinguish between the three classes. Despite the different learning performances for the studied models, which can be attributed to factors such as model size and training data volume, we can confidently state that the initial phase of the labeling process was executed effectively. This effectiveness enabled the machine learning-based method to discern patterns from the labels under study. It is important to note that the performance of the initial labeling cannot be measured using the metrics described above, as manually annotating a dataset of more than 33,000 instances would be an enormous task. The primary objective of our proposed work is to circumvent such a labor-intensive process. Note that the number of epochs varies for each model due to their differing behaviors and capacities. For instance, GPT-2, with its 1.5 billion parameters, begins to overfit the data as early as epoch 10. This necessitates an early stopping criterion to prevent overfitting and ensure optimal performance for each model. Consequently, the training duration is adjusted individually for each model to accommodate their

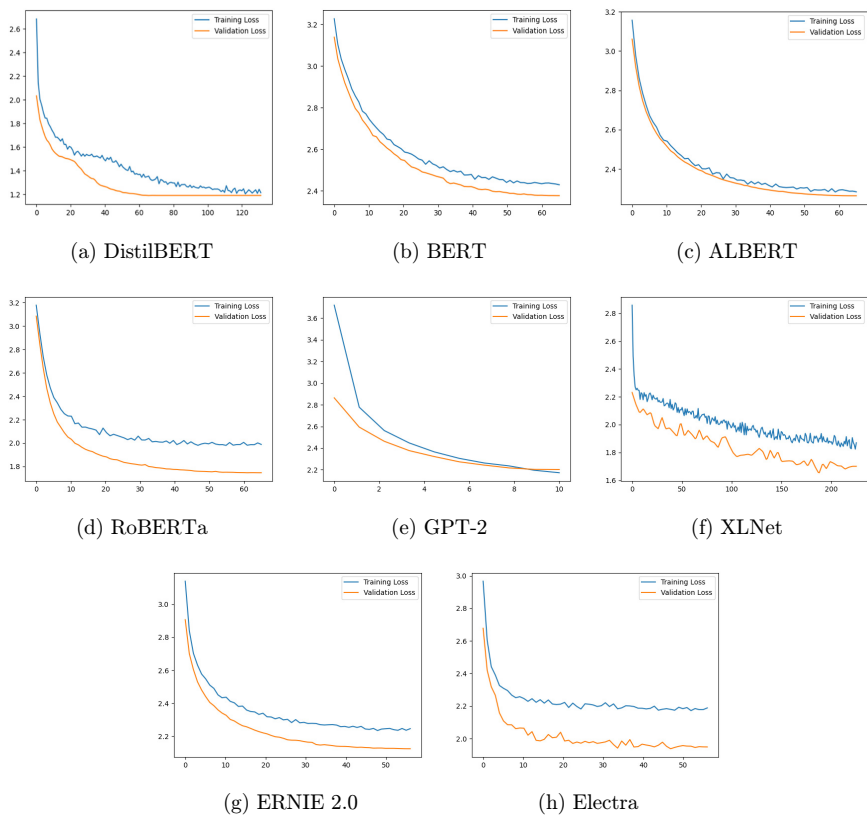


Fig. 2. Training and validation loss curves for different models.

unique characteristics and prevent overfitting while maximizing their learning efficiency. Additionally, it is worth noting that because the training library generates fractional values to track training progress, the number of epochs shown in Fig. 2, actually represents the number of tracking points saved by the trainer library.

4.4. Performance of the transformer-based models

In this section, we conducted three-class classification with labels ‘Reminder’, ‘Scheduled Activity’, and ‘General’ using transformer-based classification models, and binary classification by combining ‘Reminder’ and ‘Scheduled Activity’ into a single ‘Reminder’ label versus ‘General’ for the same models.

4.4.1. Three-class classification

We evaluated the performance of several transformer models for text classification across the studied classes: Reminder, Scheduled Activity, and General. The results, including overall accuracy, precision, recall, and F1-score, are summarized in

Table 2. Performance metrics for different models vs rule-based method.

Model	Class	Accuracy	Precision	Recall	F1-score
DistilBERT	All	0.86	0.86	0.86	0.86
	Reminder	—	0.86	0.90	0.88
	Scheduled Activity	—	0.83	0.82	0.82
	General	—	0.90	0.87	0.89
BERT	All	0.70	0.70	0.70	0.70
	Reminder	—	0.68	0.75	0.71
	Scheduled Activity	—	0.71	0.60	0.65
	General	—	0.71	0.75	0.73
ALBERT	All	0.69	0.69	0.69	0.69
	Reminder	—	0.67	0.69	0.68
	Scheduled Activity	—	0.65	0.68	0.67
	General	—	0.74	0.68	0.71
RoBERTa	All	0.79	0.79	0.79	0.79
	Reminder	—	0.81	0.86	0.83
	Scheduled Activity	—	0.74	0.75	0.74
	General	—	0.81	0.76	0.78
GPT-2	All	0.71	0.71	0.71	0.71
	Reminder	—	0.71	0.85	0.77
	Scheduled Activity	—	0.68	0.62	0.65
	General	—	0.74	0.66	0.69
Ernie 2.0	All	0.72	0.72	0.72	0.72
	Reminder	—	0.74	0.71	0.73
	Scheduled Activity	—	0.68	0.68	0.68
	General	—	0.74	0.78	0.76
Electra	All	0.74	0.74	0.74	0.74
	Reminder	—	0.68	0.87	0.76
	Scheduled Activity	—	0.72	0.65	0.68
	General	—	0.83	0.69	0.76
XLNET	All	0.79	0.79	0.79	0.79
	Reminder	—	0.77	0.92	0.84
	Scheduled Activity	—	0.81	0.72	0.60
	General	—	0.80	0.84	0.82

Table 2. Each model’s performance is broken down by individual classes as well as aggregated across all classes. For DistilBERT, the overall accuracy, precision, recall, and F1-score were all 0.86, indicating strong and balanced performance across all classes. Specifically, the precision, recall, and F1-score for the Reminder class were 0.86, 0.90, and 0.88, respectively, showing excellent capability in identifying reminders. The Scheduled Activity class achieved a precision of 0.83, recall of 0.82, and F1-score of 0.82, while the General class had scores of 0.90, 0.87, and 0.89, reflecting robust performance in general sentence classification. The BERT model showed lower overall performance compared to DistilBERT, with accuracy, precision, recall, and F1-score all at 0.70. For the Reminder class, BERT achieved a precision of 0.68, recall of 0.75, and F1-score of 0.71. The Scheduled Activity class recorded a precision of 0.71, recall of 0.60, and F1-score of 0.65. The General class had precision, recall, and F1-score of 0.71, 0.75, and 0.73, respectively. In contrast, the ALBERT model, similar in performance to BERT, had an overall accuracy,

precision, recall, and F1-score of 0.69. For the Reminder class, ALBERT achieved a precision of 0.67, recall of 0.69, and F1-score of 0.68. The Scheduled Activity class scored 0.65 for precision, 0.68 for recall, and 0.67 for F1-score. The General class had scores of 0.74 for precision, 0.68 for recall, and 0.71 for F1-score. The RoBERTa model performed better than both BERT and ALBERT, with an overall accuracy, precision, recall, and F1-score of 0.79. The Reminder class had scores of 0.81 for precision, 0.86 for recall, and 0.83 for F1-score. The Scheduled Activity class achieved 0.74 for both precision and recall, with an F1-score of 0.74. The General class showed precision, recall, and F1-score of 0.81, 0.76, and 0.78, respectively. GPT-2 also demonstrated notable performance, with overall metrics of 0.71 for accuracy, precision, recall, and F1-score. The Reminder class had high precision of 0.71, recall of 0.85, and F1-score of 0.77. The Scheduled Activity class showed lower scores with 0.68 for precision, 0.62 for recall, and 0.65 for F1-score, while the General class had precision, recall, and F1-score of 0.74, 0.66, and 0.69, respectively. In comparison to BERT, ALBERT, and GPT-2, Ernie 2.0 slightly outperformed these models, achieving an overall accuracy, precision, recall, and F1-score of 0.72 across all classes. Specifically, for the Reminder class, Ernie 2.0 yielded a precision of 0.74, recall of 0.71, and F1-score of 0.73. For the Scheduled Activity class, the precision, recall, and F1-score were all 0.68. In the General class, the model achieved precision, recall, and F1-scores of 0.74, 0.78, and 0.76, respectively. Electra demonstrated a slightly better overall performance, with an accuracy, precision, recall, and F1-score of 0.74. For the Reminder class, Electra scored 0.68 in precision, 0.87 in recall, and 0.76 in F1-score, while the Scheduled Activity class had scores of 0.72, 0.65, and 0.68, respectively. In the General class, Electra achieved 0.83 in precision, 0.69 in recall, and 0.76 in F1-score. Lastly, XLNet demonstrated a similar performance to RoBERTa, achieving an overall accuracy, precision, recall, and F1-score of 0.79. For the Reminder class, XLNet delivered a precision of 0.77, recall of 0.92, and F1-score of 0.84. The Scheduled Activity class had a precision of 0.81, recall of 0.72, and F1-score of 0.60, while the General class recorded scores of 0.80, 0.84, and 0.82 for precision, recall, and F1-score, respectively.

Analyzing each class individually is crucial as it offers insights into how well the model performs for each specific class. In this case, the model exhibits comparable performance across all classes, with F1-scores ranging from 0.60 to 0.89, which proves that there is an important correspondence between both labeling methods. The “General” class has the highest F1-score of 0.89, indicating that the model performs best on this class. This can be explained by the fact that this class presents a wide range of examples, making its classification easier compared to the other ones.

Besides, we conducted experimentation also on the chat dataset. The obtained results are given in Table 3. For DistilBERT, the model achieved the highest overall performance with an accuracy, precision, recall, and F1-score of 0.86 for the combined classes. Specifically, the precision, recall, and F1-score for the Reminder class were 0.77, 0.85, and 0.81, respectively, indicating strong performance in identifying reminders. For the Scheduled Activity class, the scores were significantly lower, with

Table 3. Performance metrics on the chat dataset for classification models.

Model	Class	Accuracy	Precision	Recall	F1-score
DistilBERT	All	0.86	0.86	0.86	0.86
	Reminder	—	0.77	0.85	0.81
	Scheduled Activity	—	0.19	0.19	0.19
	General	—	0.97	0.92	0.95
BERT	All	0.55	0.55	0.55	0.55
	Reminder	—	0.39	0.81	0.53
	Scheduled Activity	—	0.04	0.06	0.05
	General	—	1.00	0.47	0.68
ALBERT	All	0.72	0.72	0.72	0.72
	Reminder	—	0.56	0.74	0.64
	Scheduled Activity	—	0.32	0.31	0.26
	General	—	0.91	0.74	0.82
RoBERTa	All	0.71	0.71	0.71	0.71
	Reminder	—	0.52	0.86	0.65
	Scheduled Activity	—	0.30	0.44	0.36
	General	—	0.96	0.66	0.78
GPT-2	All	0.36	0.36	0.36	0.36
	Reminder	—	0.31	0.46	0.37
	Scheduled Activity	—	0.11	0.81	0.20
	General	—	0.90	0.28	0.43
Ernie 2.0	All	0.72	0.72	0.72	0.72
	Reminder	—	0.74	0.71	0.73
	Scheduled Activity	—	0.68	0.68	0.68
	General	—	0.74	0.78	0.76
Electra	All	0.74	0.75	0.74	0.74
	Reminder	—	0.68	0.87	0.76
	Scheduled Activity	—	0.72	0.65	0.68
	General	—	0.83	0.69	0.76
XLNET	All	0.51	0.51	0.51	0.51
	Reminder	—	0.39	0.73	0.51
	Scheduled Activity	—	0.07	0.19	0.10
	General	—	0.91	0.43	0.59

precision, recall, and F1-score all at 0.19. This weak performance can be attributed to the limited number of instances (only 16) in this class within the chat dataset. Additionally, each sentence in this class is an imperative sentence, such as “Set up a meeting for tomorrow” which could easily be mistaken for a task to do, thereby complicating accurate classification. In the General class, DistilBERT excelled with a precision of 0.97, recall of 0.92, and F1-score of 0.95, showing robust performance across all metrics. BERT’s performance was significantly lower than DistilBERT’s, with an overall accuracy, precision, recall, and F1-score of 0.55. Within the Reminder class, BERT achieved a precision of 0.39, recall of 0.81, and F1-score of 0.53. The Scheduled Activity class recorded the weakest performance with a precision of 0.04, recall of 0.06, and F1-score of 0.05. The General class showed moderate results with scores of 1.00, 0.47, and 0.68 for precision, recall, and F1-score, respectively. ALBERT’s overall performance was similar to BERT’s with an overall score of 0.72 across all metrics. For the Reminder class, ALBERT achieved a precision of 0.56,

recall of 0.74, and F1-score of 0.64. The Scheduled Activity class scores were 0.32, 0.31, and 0.26 for precision, recall, F1-score, respectively, while 0.91 of precision, 0.74 of recall, and 0.82 of F1-score were obtained for General class. RoBERTa performed better than BERT but still trailed DistilBERT, achieving an overall score of 0.71. For the Reminder class, RoBERTa's precision, recall, and F1-score were 0.52, 0.86, and 0.65, respectively. The Scheduled Activity class had scores of 0.30, 0.44, and 0.36 for precision, recall, and F1-score, respectively. For the General class, RoBERTa scored 0.96, 0.66, and 0.78 for precision, recall, and F1-score, respectively.

Lastly, GPT-2 had an overall performance of 0.36, similar to BERT but with slightly different class-specific results. The Reminder class showed a precision of 0.31, recall of 0.46, and F1-score of 0.37. The Scheduled Activity class recorded a precision of 0.11, recall of 0.81, and F1-score of 0.20, while the General class had scores of 0.90, 0.28, and 0.43 for precision, recall and F1-score, respectively. Ernie 2.0 performed comparably to ALBERT and slightly outperformed BERT, achieving an overall accuracy, precision, recall, and F1-score of 0.72. For the Reminder class, Ernie 2.0 achieved a precision of 0.74, recall of 0.71, and F1-score of 0.73. In the Scheduled Activity class, the precision, recall, and F1-score were all 0.68, reflecting an acceptable performance. In the General class, Ernie 2.0 achieved a precision of 0.74, recall of 0.78, and F1-score of 0.76. Electra demonstrated a slightly better performance than Ernie 2.0, with an overall score of 0.74. For the Reminder class, Electra achieved a precision of 0.68, recall of 0.87, and F1-score of 0.76. The Scheduled Activity class had scores of 0.72, 0.65, and 0.68 for precision, recall, and F1-score, respectively. Finally, the General class achieved a precision, recall and F1-score of 0.83, 0.69, and 0.76, respectively. Lastly, XLNet performed similarly to BERT, with an overall score of 0.51. In the Reminder class, XLNet achieved a precision of 0.39, recall of 0.73, and F1-score of 0.51. The Scheduled Activity class recorded low scores with precision, recall, and F1-score of 0.07, 0.19, and 0.10, respectively. For the General class, XLNet showed better performance, with scores of 0.91 of precision, 0.43 of recall, and 0.59 of F1-score.

4.4.2. Binary classification

For binary classification, we considered a setup that merges both the Reminder and Scheduled Activity classes because both represent tasks, either with or without a specific time schedule. The results, summarized in Table 4, show that across different models, the merged task class (combining Reminder and Scheduled Activity) and General class demonstrated varied performance. In fact, DistilBERT achieved the highest overall performance with an accuracy of 0.88, precision of 0.88, recall of 0.88, and F1-score of 0.88 for the combined classes. For the Reminder class, the precision, recall, and F1-score were 0.89, 0.89, and 0.89, respectively. The General class also performed well, with precision, recall, and F1-score of 0.86, 0.87, and 0.87, respectively. BERT followed with an overall accuracy of 0.78, precision of 0.78, recall of 0.78, and F1-score of 0.78. For the Reminder class, BERT achieved a precision of

Table 4. Performance results for different models vs rule-based method.

Model	Class	Accuracy	Precision	Recall	F1-score
DistilBERT	All	0.88	0.88	0.88	0.88
	Reminder	—	0.89	0.89	0.89
	General	—	0.86	0.87	0.87
BERT	All	0.78	0.78	0.78	0.78
	Reminder	—	0.82	0.75	0.79
	General	—	0.73	0.80	0.76
ALBERT	All	0.79	0.79	0.79	0.79
	Reminder	—	0.80	0.83	0.81
	General	—	0.78	0.75	0.76
RoBERTa	All	0.82	0.82	0.82	0.82
	Reminder	—	0.84	0.84	0.84
	General	—	0.80	0.80	0.80
GPT-2	All	0.77	0.77	0.77	0.77
	Reminder	—	0.78	0.75	0.76
	General	—	0.76	0.78	0.77
Ernie 2.0	All	0.79	0.79	0.79	0.79
	Reminder	—	0.82	0.80	0.81
	General	—	0.76	0.78	0.77
Electra	All	0.80	0.80	0.80	0.80
	Reminder	—	0.82	0.80	0.81
	General	—	0.77	0.79	0.78
XLNET	All	0.63	0.73	0.63	0.65
	Reminder	—	0.60	0.98	0.74
	General	—	0.90	0.20	0.33

0.82, recall of 0.75, and F1-score of 0.79. In the General class, BERT had precision, recall, and F1-score of 0.73, 0.80, and 0.76, respectively. ALBERT’s overall performance was close to BERT’s, with an accuracy of 0.79, precision of 0.79, recall of 0.79, and F1-score of 0.79. The Reminder class achieved precision, recall, and F1-score of 0.80, 0.83, and 0.81, while the General class scored 0.78, 0.75, and 0.76, respectively. RoBERTa slightly outperformed BERT, obtaining an overall accuracy of 0.82, precision of 0.82, recall of 0.82, and F1-score of 0.82. The Reminder class had scores of 0.84, 0.84, and 0.84 for precision, recall, and F1-score, respectively, while the General class scored 0.80, 0.80, and 0.80. GPT-2’s overall performance was lower, with an accuracy of 0.77, precision of 0.77, recall of 0.77, and F1-score of 0.77. For the Reminder class, GPT-2 recorded scores of 0.78, 0.75, and 0.76 for precision, recall, and F1-score, while the General class had 0.76, 0.78, and 0.77 for the same metrics. Ernie 2.0 had an overall accuracy of 0.79, precision of 0.79, recall of 0.79, and F1-score of 0.79. The Reminder class showed a precision, recall, and F1-score of 0.82, 0.80, and 0.81, while the General class achieved 0.76, 0.78, and 0.77 for these metrics. Electra achieved an overall accuracy of 0.80, precision of 0.80, recall of 0.80, and F1-score of 0.80. The Reminder class had scores of 0.82, 0.80, and 0.81 for precision, recall, and F1-score, respectively, while the General class achieved 0.77, 0.79, and 0.78.

Table 5. Performance results for different models on chat dataset.

Model	Class	Accuracy	Precision	Recall	F1-score
DistilBERT	All	0.92	0.92	0.92	0.92
	Reminder	—	0.87	0.90	0.88
	General	—	0.95	0.94	0.93
BERT	All	0.81	0.81	0.81	0.81
	Reminder	—	0.65	0.97	0.78
	General	—	0.98	0.73	0.83
ALBERT	All	0.71	0.71	0.71	0.71
	Reminder	—	0.57	0.63	0.60
	General	—	0.80	0.75	0.77
RoBERTa	All	0.83	0.83	0.83	0.83
	Reminder	—	0.70	0.87	0.77
	General	—	0.92	0.80	0.86
GPT-2	All	0.70	0.70	0.70	0.70
	Reminder	—	0.54	0.94	0.68
	General	—	0.95	0.58	0.72
Ernie 2.0	All	0.75	0.75	0.75	0.75
	Reminder	—	0.60	0.80	0.69
	General	—	0.88	0.72	0.79
Electra	All	0.77	0.77	0.77	0.77
	Reminder	—	0.63	0.80	0.71
	General	—	0.88	0.76	0.82
XLNET	All	0.67	0.67	0.67	0.67
	Reminder	—	0.51	0.79	0.67
	General	—	0.85	0.61	0.71

For the chat dataset, Table 5 presents the performance of various models across different metrics. The results highlight substantial variation in accuracy, precision, recall, and F1-score between the “Reminder” and “General” classes. DistilBERT stood out with the highest overall accuracy of 0.92. Specifically, for the “Reminder” class, it achieved a precision of 0.87, recall of 0.90, and F1-score of 0.88. The “General” class showed even stronger results, with precision, recall, and F1-scores of 0.95, 0.94, and 0.93, respectively. BERT followed with a slightly lower overall accuracy of 0.81. For the “Reminder” class, it delivered precision of 0.65, recall of 0.97, and an F1-score of 0.78. However, its performance for the “General” class was considerably better, with a precision of 0.98, recall of 0.73, and F1-score of 0.83. ALBERT lagged behind in terms of overall accuracy at 0.71. In the “Reminder” class, its precision was 0.57, recall was 0.63, and F1-score was 0.60, while its performance for the “General” class showed better results with precision of 0.80, recall of 0.75, and F1-score of 0.77. RoBERTa performed quite well, achieving an overall accuracy of 0.83. It attained a precision of 0.70, recall of 0.87, and F1-score of 0.77 for the “Reminder” class. The “General” class, however, scored higher with precision of 0.92, recall of 0.80, and F1-score of 0.86. GPT-2’s overall accuracy was 0.70, with its “Reminder” class results showing a precision of 0.54, recall of 0.94, and F1-score of 0.68. For the “General” class, GPT-2’s performance improved with a precision of 0.95, recall of 0.58, and F1-score of 0.72. Ernie 2.0 yielded an overall accuracy of 0.75.

For the “Reminder” class, it reached precision of 0.60, recall of 0.80, and F1-score of 0.69. The “General” class metrics were slightly higher with precision of 0.88, recall of 0.72, and F1-score of 0.79. Electra displayed an overall accuracy of 0.77, with the “Reminder” class posting precision of 0.63, recall of 0.80, and F1-score of 0.71. Its “General” class performance was stronger, with precision, recall, and F1-score of 0.88, 0.76, and 0.82, respectively. Finally, XLNET had the lowest overall accuracy of 0.67. Its “Reminder” class results were modest, with precision of 0.51, recall of 0.79, and F1-score of 0.67, while the “General” class recorded precision of 0.85, recall of 0.61, and F1-score of 0.71.

4.5. Performance comparison with generative AI models

In this section, we report three-class and binary classifications of the proposed method in comparison to both zero-shot and one-shot strategies using generative AI models.

4.5.1. Three-class classification

We conducted experimentation also on the chat dataset with various GPT-based models along with LLaMA-3 70B. The obtained results are given in Table 6. The models were prompted with the following instructions: “You are an intelligent classifier. Your task is to classify text into three categories: ‘Scheduled Activity’, ‘Reminder’, and ‘General’. The ‘Reminder’ category includes tasks that need to be

Table 6. Performance metrics on the chat dataset for DistilBERT vs generative AI models using zero-shot learning.

Model	Class	Accuracy	Precision	Recall	F1-score
DistilBERT	All	0.86	0.86	0.86	0.86
	Reminder	—	0.77	0.85	0.81
	Scheduled Activity	—	0.19	0.19	0.19
	General	—	0.97	0.92	0.95
GPT-4	All	0.67	0.67	0.67	0.67
	Reminder	—	0.33	0.01	0.02
	Scheduled Activity	—	0.36	0.81	0.50
	General	—	0.71	0.94	0.81
GPT-3.5-turbo-0613	All	0.70	0.70	0.70	0.70
	Reminder	—	0.64	0.08	0.14
	Scheduled Activity	—	0.44	0.75	0.56
	General	—	0.73	0.96	0.83
GPT-4o	All	0.73	0.73	0.73	0.73
	Reminder	—	0.65	0.15	0.25
	Scheduled Activity	—	0.57	0.50	0.53
	General	—	0.74	1.00	0.85
LLaMA-3 70B	All	0.71	0.71	0.71	0.71
	Reminder	—	0.54	0.51	0.53
	Scheduled Activity	—	0.45	0.94	0.61
	General	—	0.83	0.78	0.80

accomplished, which may not necessarily have a specific time frame. ‘Scheduled Activity’ refers to time-bound events such as meetings, appointments, and calls that occur at a predetermined date and time. The ‘General’ category includes sentences that do not fit into the specific categories of reminders or scheduled activities. Each sentence is presented in the format *sentence_i: sentence*. Your output should be in the format of *sentence_i: label*. These are the sentences to be labeled.” The temperature was set to zero to ensure deterministic outputs. For generative-AI models, the results varied significantly:

- GPT-4: The overall performance was lower with an accuracy of 0.67. While it performed reasonably well on the General class (precision 0.71, recall 0.94, F1-score 0.81), it struggled with the Reminder class (precision 0.33, recall 0.01, F1-score 0.02). The Scheduled Activity class had moderate performance (precision 0.36, recall 0.81, F1-score 0.50).
- GPT-3.5-turbo-0613: This model had an overall accuracy of 0.70. It performed best in the General class (precision 0.73, recall 0.96, F1-score 0.83), but poorly in the Reminder class (precision 0.64, recall 0.08, F1-score 0.14). The Scheduled Activity class showed moderate performance (precision 0.44, recall 0.75, F1-score 0.56).
- GPT-4o: This variant achieved an overall accuracy of 0.73. It had good performance in the General class (precision 0.74, recall 1.00, F1-score 0.85), but showed limited effectiveness in the Reminder class (precision 0.65, recall 0.15, F1-score 0.25). The Scheduled Activity class had moderate performance (precision 0.57, recall 0.50, F1-score 0.53).
- LLaMA-3 70b: This model demonstrated an overall accuracy of 0.71, with varying performance across classes. In the ‘General’ class, it achieved a precision of 0.83, recall of 0.78, and F1-score of 0.80, indicating strong balanced performance. In contrast, for the ‘Reminder’ class, its precision and recall were more moderate at 0.54 and 0.51, respectively, leading to an F1-score of 0.53. The model’s highest recall was in the ‘Scheduled Activity’ class, where it reached 0.94, though precision was comparatively lower at 0.45, resulting in an F1-score of 0.61. This pattern suggests that while LLaMA-3 excels in capturing relevant instances in the ‘Scheduled Activity’ class, it may also produce a higher number of false positives.

These results indicate that despite the larger model sizes and advanced capabilities of generative-AI models, DistilBERT, a smaller and more efficient model, outperformed them in text classification tasks. The variability in generative-AI models’ performance underscores the dependency on user understanding and prompt engineering. Furthermore, our model demonstrated better performance compared to using generative AI models for zero-shot prompt classification. This highlights the efficiency of our hybrid approach that integrates rule-based annotation with a fine-tuned, distilled transformer model, providing a cost-effective and robust solution for text classification.

Table 7. Performance metrics on the chat dataset for DistilBERT vs generative AI models using one-shot learning.

Model	Class	Accuracy	Precision	Recall	F1-score
DistilBERT	All	0.86	0.86	0.86	0.86
	Reminder	—	0.77	0.85	0.81
	Scheduled Activity	—	0.19	0.19	0.19
	General	—	0.97	0.92	0.95
GPT-4	All	0.67	0.67	0.67	0.67
	Reminder	—	0.36	0.81	0.50
	Scheduled Activity	—	0.36	0.81	0.50
	General	—	0.71	0.94	0.81
GPT-3.5-turbo-0613	All	0.70	0.70	0.70	0.70
	Reminder	—	0.50	0.08	0.14
	Scheduled Activity	—	0.50	0.62	0.56
	General	—	0.72	0.97	0.83
GPT-4o	All	0.72	0.72	0.72	0.72
	Reminder	—	0.62	0.09	0.16
	Scheduled Activity	—	0.60	0.56	0.58
	General	—	0.73	1.00	0.84
LLaMA-3 70B	All	0.73	0.73	0.73	0.73
	Reminder	—	0.57	0.55	0.56
	Scheduled Activity	—	0.48	0.94	0.64
	General	—	0.84	0.79	0.64

We conducted additional experimentation on the chat dataset using one-shot learning. The results are presented in Table 7. Each model was prompted with the following instructions: “*You are an intelligent classifier. Your task is to classify text into three categories: ‘Scheduled Activity,’ ‘Reminder,’ and ‘General.’ The ‘Reminder’ category includes tasks that need to be accomplished but may not have a specific time frame. ‘Scheduled Activity’ refers to time-bound events such as meetings, appointments, and calls scheduled at a specific date and time. The ‘General’ category includes sentences that do not fit into the specific categories of reminders or scheduled activities. Each sentence is presented in the format sentence_i: sentence. Your output should be in the format sentence_i: label. These are the sentences to be labeled: Example: Sentences: Please call if you have any questions. Some items may not come back at the same price. Expected Output: [‘sentence_1’: ‘Reminder’, ‘sentence_2’: ‘General’].*” The temperature was set to zero to ensure deterministic outputs. For each of the generative-AI models, generally, results demonstrated good performances:

- GPT-4: Achieved an overall accuracy of 0.67, with the best performance in the General class (precision 0.71, recall 0.94, F1-score 0.81). It struggled in the Reminder class (precision 0.36, recall 0.81, F1-score 0.50) and showed moderate performance in Scheduled Activity (precision 0.36, recall 0.81, F1-score 0.50).
- GPT-3.5-turbo-0613: This model scored an overall accuracy of 0.70, with strong performance in the General class (precision 0.72, recall 0.97, F1-score 0.83).

However, it underperformed in Reminder (precision 0.50, recall 0.08, F1-score 0.14) and had moderate results for Scheduled Activity (precision 0.50, recall 0.62, F1-score 0.56).

- GPT-4o: An overall accuracy of 0.72 was achieved. It excelled in the General class (precision 0.73, recall 1.00, F1-score 0.84) but showed limited effectiveness in the Reminder class (precision 0.62, recall 0.09, F1-score 0.16) and moderate performance in Scheduled Activity (precision 0.60, recall 0.56, F1-score 0.58).
- LLaMA-3 70B: This model demonstrated an overall accuracy of 0.73, showing strong balanced performance in the General class (precision 0.84, recall 0.79, F1-score 0.64) and moderate results in Reminder (precision 0.57, recall 0.55, F1-score 0.56). Its highest recall was in Scheduled Activity, where it reached 0.94 (precision 0.48, F1-score 0.64), suggesting effectiveness in capturing scheduled instances but with some false positives.

4.5.2. Binary classification

In the zero-shot configuration, the prompt provided only the task instructions, without any labeled examples, allowing us to evaluate each model's innate ability to generalize to the classification task without prior context. In fact, the used instructions are as follows: “*You are an intelligent classifier. Your task is to classify text into two categories: ‘Reminder,’ and ‘General.’ The ‘Reminder’ category includes tasks that need to be accomplished. It also includes time-bound events such as meetings, appointments, and calls that occur at a predetermined date and time. The ‘General’ category includes sentences that do not fit into the specific category of reminder. Each sentence is presented in the format sentence_i: sentence. Your output should be in the format sentence_i: label. These are the sentences to be labeled:.*” The recorded results are given in Table 8 showing that the proposed method has a better performance in comparison to GPT-based models along with LLaMA-3 70B:

- DistilBERT: This model achieved the highest overall accuracy of 0.92, with balanced precision, recall, and F1-scores across both “Reminder” (precision 0.87, recall 0.90, F1-score 0.88) and “General” (precision 0.95, recall 0.94, F1-score 0.93) classes. This indicates that DistilBERT effectively distinguishes between the two categories outperforming the large generative models.
- GPT-4 and GPT-4o: Both models achieved an overall accuracy of 0.74. In the “Reminder” category, they displayed high precision (1.00) but limited recall (0.23), resulting in a lower F1-score of 0.37. In contrast, their performance in the “General” class was strong, with recall scores of 1.00 and F1-scores of 0.83. This precision-recall imbalance suggests that GPT-4 and GPT-4o tend to over-identify statements as non-reminders, potentially overlooking some “Reminder” instances.
- GPT-3.5-turbo-0613: This variant scored an overall accuracy of 0.75, GPT-3.5-turbo demonstrated strong precision in the “Reminder” category (0.88) but low recall (0.29), indicating a similar trend to GPT-4 in missing some reminder-related

Table 8. Performance metrics on the chat dataset for DistilBERT vs generative AI models using zero-shot learning.

Model	Class	Accuracy	Precision	Recall	F1-score
DistilBERT	All	0.92	0.92	0.92	0.92
	Reminder	—	0.87	0.90	0.88
	General	—	0.95	0.94	0.93
GPT-4	All	0.74	0.74	0.74	0.74
	Reminder	—	1.00	0.23	0.37
	General	—	0.71	1.00	0.83
GPT-3.5-turbo-0613	All	0.75	0.75	0.75	0.75
	Reminder	—	0.88	0.29	0.44
	General	—	0.73	0.98	0.84
GPT-4o	All	0.74	0.74	0.74	0.74
	Reminder	—	1.00	0.23	0.37
	General	—	0.71	1.00	0.83
LLaMA-3 70B	All	0.75	0.75	0.75	0.75
	Reminder	—	0.74	0.41	0.53
	General	—	0.75	0.92	0.83

instances. Its performance in the “General” category was strong, with an F1-score of 0.84.

- LLaMA-3 70B: This model attained an accuracy of 0.75 and exhibited a better balance between precision and recall in the “Reminder” class, with an F1-score of 0.53. Its “General” category performance was robust, showing precision of 0.75 and recall of 0.92. LLaMA-3 suggest that it has a higher capacity for capturing both reminder and general instances than the other generative models in zero-shot conditions.

In the one-shot configuration, the instructions were similar to those used in the zero-shot setup, with the addition of a single labeled example in the prompt, providing each model with contextual guidance on the classification task. Thus, the final instructions are given as follows: “*You are an intelligent classifier. Your task is to classify text into two categories: ‘Reminder,’ and ‘General.’ The ‘Reminder’ category includes tasks that need to be accomplished. It also includes time-bound events such as meetings, appointments, and calls that occur at a predetermined date and time. The ‘General’ category includes sentences that do not fit into the specific category of reminder. Each sentence is presented in the format sentence_i: sentence. Your output should be in the format sentence_i: label. These are the sentences to be labeled: Example: Sentences: Please call if you have any questions. Some items may not come back at the same price. Expected Output: [‘sentence_1’: ‘Reminder’, ‘sentence_2’: ‘General’].*” Thus, in comparison to DistilBERT performance, GPT-based models and LLaMA-3 70B recorded a lower performances (Table 9):

- GPT-4: Performance dipped slightly in one-shot learning, with an accuracy of 0.73. In the “Reminder” category, precision was 0.96 but recall remained low

Table 9. Performance metrics on the chat dataset for DistilBERT vs generative AI models using one-shot learning.

Model	Class	Accuracy	Precision	Recall	F1-score
DistilBERT	All	0.92	0.92	0.92	0.92
	Reminder	—	0.87	0.90	0.88
	General	—	0.95	0.94	0.93
GPT-4	All	0.73	0.73	0.73	0.73
	Reminder	—	0.96	0.23	0.37
	General	—	0.71	0.99	0.83
GPT-3.5-turbo-0613	All	0.73	0.73	0.73	0.73
	Reminder	—	0.92	0.24	0.38
	General	—	0.72	0.99	0.83
GPT-4o	All	0.74	0.74	0.74	0.74
	Reminder	—	1.00	0.23	0.37
	General	—	0.71	1.00	0.83
LLaMA-3 70B	All	0.75	0.75	0.75	0.75
	Reminder	—	0.74	0.41	0.53
	General	—	0.75	0.92	0.83

- (0.23), resulting in an F1-score of 0.37. For the “General” category, GPT-4 achieved a recall of 0.99 with an F1-score of 0.83.
- GPT-3.5-turbo-0613: In the one-shot setup, GPT-3.5-turbo achieved an accuracy of 0.73, showing marginal improvements in recall for the “Reminder” class (0.24), while maintaining high precision. Its F1-score for “General” remained strong at 0.83, similar to its zero-shot performance.
 - GPT-4o: the model achieved an accuracy of 0.74. The “Reminder” class had high precision (1.00) but low recall (0.23), giving it an F1-score of 0.37. In the “General” category, it continued to perform strongly with a recall of 1.00 and F1-score of 0.83.
 - LLaMA-3 70B: LLaMA-3 70B achieved a balanced overall accuracy of 0.75, demonstrating modest improvement in recall within the “Reminder” class (0.41), leading to an F1-score of 0.53. In the “General” class, it performed well, maintaining a high precision of 0.75 and recall of 0.92.

5. Discussion

In our study, we investigated the performance of various transformer models for text classification using rule-based annotated data. Our analysis highlights the importance of model size and the benefits of model distillation, particularly with DistilBERT outperforming larger models such as BERT, RoBERTa, GPT-2, and models like Ernie 2.0, XLNet, and Electra as well as models with fewer parameters like ALBERT.

- The performance of transformer models in text classification tasks is influenced significantly by their size. Larger models, such as BERT and RoBERTa, have

more parameters and can capture complex patterns in data. However, this increased capacity often comes with higher computational costs and a greater tendency to overfit, particularly when the training data is limited or when early stopping is not effectively utilized. Ernie 2.0 and Electra, although designed to efficiently handle specific downstream tasks, did not surpass DistilBERT's balanced performance across both the "Reminder" and "General" classes. For instance, Ernie 2.0 and Electra achieved only modest gains over BERT, with slight improvements in precision and recall but did not match DistilBERT's higher F1-scores and computational efficiency. XLNet, with its unique permutation-based architecture, struggled with lower accuracy and recall, particularly in the "General" class, highlighting limitations in its robustness for this classification task. For instance, some of these models were prone to overfitting as indicated by their training loss curves and classification metrics. In contrast, DistilBERT, with its 66 million parameters, strikes a balance between performance and efficiency. As a distilled version of BERT, it retains approximately from 95% to 97% of BERT's language understanding while having 60% fewer parameters. This reduction in size results in faster training and inference times, as well as lower computational costs, making it particularly suited for scenarios where computational resources are constrained.

- DistilBERT's success can be attributed to the process of model distillation, which involves transferring knowledge from a larger, more complex teacher model (BERT) to a smaller, more efficient student model (DistilBERT). This process involves training the smaller model to mimic the behavior of the larger model by learning from its predictions and internal representations. The result is a model that is not only more compact but also retains a high level of performance on various tasks. The effectiveness of DistilBERT can be partly explained by its ability to leverage the pre-trained knowledge of BERT while reducing the number of parameters. Distillation helps in preserving the crucial aspects of the teacher model's knowledge while discarding redundant information, thus achieving a good trade-off between efficiency and performance. This allows DistilBERT to perform well even with a fixed dataset size of around 33,000 instances, which might be suboptimal for larger models prone to overfitting.

In our comparative analysis, BERT, despite its higher parameter count of 110 million, did not outperform DistilBERT on our dataset. BERT's increased complexity may lead to overfitting, especially if early stopping is not well-tuned, resulting in diminished classification performance. RoBERTa, particularly the large variant with 355 million parameters, along with Ernie 2.0, XLENT and Electra also fell short compared to DistilBERT. The higher computational demands and tendency to overfit may not yield proportional gains in classification accuracy. Although ALBERT's base version has fewer parameters (11 million), its performance was still inferior to that of DistilBERT. ALBERT's parameter reduction approach, which involves sharing weights across layers and factorized embedding parameterization,

may not always be optimal for specific classification tasks. Lastly, GPT-2, designed primarily for text generation, showed less effectiveness in classification tasks compared to DistilBERT, highlighting its architectural optimization for generative rather than discriminative tasks. In what concerns prompt engineering using LLMs such as LLaMA-3 70B and GPT-4o, one of OpenAI's largest models, our annotation method showed prominent results compared to prompting LLMs with zero shot strategy. In fact, this could not reflect the real performance of generative AI models since it is strongly dependent on the quality of the prompt and the used instructions.

6. Conclusion

In this paper, we investigated the efficacy of combining rule-based methods with various pretrained transformer models, including DistilBERT, for text classification. Our findings reveal that DistilBERT, despite its smaller size, outperformed larger models, highlighting a critical trade-off between model size and training data requirements. The superior performance of DistilBERT can be attributed to its efficiency in learning from well-annotated data and the effective application of regular expressions for initial data labeling. The comparison across different models demonstrated that larger models do not necessarily guarantee better performance in specific classification tasks, particularly when the training data is limited or highly specialized. The success of DistilBERT emphasizes the importance of choosing the right model size to balance computational efficiency and accuracy, especially in practical applications where resources might be constrained. Our experiments also underscored the significant role of user understanding in the labeling process. This dependency suggests that future research should focus on developing multi-label learning techniques to mitigate ambiguities and improve the model's ability to handle complex and overlapping annotations.³⁷ Such approaches could enhance the model's robustness and accuracy in diverse real-world scenarios. Additionally, replacing rule-based expressions with well-defined instructions for annotation could be interesting to test, especially given the rapid advancements in generative AI models.

ORCID

Arafet Sbei  <https://orcid.org/0009-0008-5887-1569>

Khaoula ElBedoui  <https://orcid.org/0000-0001-9687-720X>

Walid Barhoumi  <https://orcid.org/0000-0003-2123-4992>

References

1. R. Ghanem and H. Erbay, Spam detection on social networks using deep contextualized word representation, *Multimed. Tools Appl.* **82** (2023) 3697–3712.
2. S. Rao, A. K. Verma and T. Bhatia, Hybrid ensemble framework with self-attention mechanism for social spam detection on imbalanced data, *Expert Syst. Appl.* **217** (2023) 119594.

3. C. Liu and X. Xu, AMFF: A new attention-based multi-feature fusion method for intention recognition, *Knowl.-Based Syst.* **233** (2021) 107525.
4. R. Wagh and P. Punde, Survey on sentiment analysis using twitter dataset, in *2018 Second Int. Conf. Electronics, Communication and Aerospace Technology (ICECA)* (IEEE, 2018), pp. 208–211.
5. Q. Li, H. Peng, J. Li, C. Xia, R. Yang, L. Sun and L. He, A survey on text classification: From traditional to deep learning, *ACM Trans. Intell. Syst. Technol.* **13**(2) (2022) 1–41.
6. Y. Meng, Y. Zhang, J. Huang, C. Xiong, H. Ji, C. Zhang and J. Han, Text classification using label names only: A language model self-training approach, in *Proc. 2020 Conf. Empirical Methods in Natural Language Processing (EMNLP)* (Association for Computational Linguistics, 2020), pp. 9006–9017.
7. F. J  nez-Martino, E. Fidalgo, S. Gonz  -Mart  nez and J. Velasco-Mata, Classification of spam emails through hierarchical clustering and supervised learning, preprint (2020), arXiv:2005.08773.
8. A. Sbei, K. ElBedoui and W. Barhoumi, Synergistic text annotation based on rule-based expressions and DistilBERT, in *Asian Conf. Intelligent Information and Database Systems* (Springer Nature Singapore, Singapore, 2024), pp. 408–422.
9. J. Shetty and J. Adibi, The Enron email dataset database schema and brief statistical report, Information Sciences Institute Technical Report, University of Southern California (2004), pp. 120–128.
10. M. Neves and J.   va, An extensive review of tools for manual annotation of documents, *Brief. Bioinform.* **22**(1) (2021) 146–163.
11. M. van Gompel and M. Reynaert, FoLiA: A practical XML format for linguistic annotation—a descriptive and comparative study, *Comput. Linguist. Neth. J.* **3** (2013) 63–81.
12. R. Islamaj, D. Kwon, S. Kim and Z. Lu, TeamTat: A collaborative text annotation tool, *Nucleic Acids Res.* **48**(W1) (2020) W5–W11.
13. Z. Zhang, E. Strubell and E. Hovy, A survey of active learning for natural language processing, in *Proc. 2022 Conf. Empirical Methods in Natural Language Processing (EMNLP)* (Association for Computational Linguistics, 2022), pp. 6166–6190.
14. J. M. Cejuela, P. McQuilton, L. Ponting, S. J. Marygold, R. Stefancsik, G. H. Millburn, B. Rost and FlyBase Consortium, tagtog: Interactive and text-mining-assisted annotation of gene mentions in PLOS full-text articles, *Database* **2014** (2014) bau033.
15. C. Schroder, A. Niekler and M. Potthast, Revisiting uncertainty-based query strategies for active learning with transformers, in *Findings of the Association for Computational Linguistics (ACL)* (Association for Computational Linguistics, 2022), pp. 2194–2203.
16. X. Liu, G. Zhou, M. Kong, Z. Yin, X. Li, L. Yin and W. Zheng, Developing multi-labelled corpus of twitter short texts: A semi-automatic method, *Systems* **11**(8) (2023) 390.
17. S. Saifullah, R. Dre  wski, F. A. Dwiyanto, A. S. Aribowo, Y. Fauziah and N. H. Cahyana, Automated text annotation using a semi-supervised approach with meta vectorizer and machine learning algorithms for hate speech detection, *Appl. Sci.* **14**(3) (2024) 1078.
18. H. Rouzegar and M. Makrehchi, Enhancing text classification through LLM-driven active learning and human annotation, preprint (2024), arXiv:2406.12114.
19. A. Karim, S. Azam, B. Shanmugam and K. Kannoorpatti, Efficient clustering of emails into spam and ham: The foundational study of a comprehensive unsupervised framework, *IEEE Access* **8** (2020) 154759–154788.
20. S. H. R. Rasmussen, A. Bor, M. Osmundsen and M. B. Petersen, ‘Super-unsupervised’ classification for labelling text: Online political hostility as an illustration, *Br. J. Polit. Sci.* **54**(1) (2024) 179–200.

21. W. W. Chapman, W. Bridewell, P. Hanbury, G. F. Cooper and B. G. Buchanan, A simple algorithm for identifying negated findings and diseases in discharge summaries, *J. Biomed. Inform.* **34**(5) (2001) 301–310.
22. A. Smit, S. Jain, P. Rajpurkar, A. Pareek, A. Y. Ng and M. P. Lungren, CheXbert: Combining automatic labelers and expert annotations for accurate radiology report labeling using BERT, preprint (2020), arXiv:2004.09167.
23. R. Zhong, K. Lee, Z. Zhang and D. Klein, Adapting language models for zero-shot learning by meta-tuning on dataset and prompt collections, preprint (2021), arXiv:2104.04670.
24. Y. Ma, S. Luo, Y. M. Shang, Y. Zhang and Z. Li, CodePrompt: Improving source code-related classification with knowledge features through prompt learning, preprint (2024), arXiv:2401.05544.
25. L. Caruccio, S. Cirillo, G. Polese, G. Solimando, S. Sundaramurthy and G. Tortora, Claude 2.0 large language model: Tackling a real-world classification problem with a new iterative prompt engineering approach, *Intell. Syst. Appl.* **21** (2024) 200336.
26. N. Pangakis and S. Wolken, Knowledge distillation in automated annotation: Supervised text classification with LLM-generated training labels, preprint (2024), arXiv:2406.17633.
27. S. Bird, E. Klein and E. Loper, *Natural Language Processing with Python: Analyzing Text with the Natural Language Toolkit* (O'Reilly Media, 2009).
28. J. Devlin, M. W. Chang, K. Lee and K. Toutanova, Bert: Pre-training of deep bidirectional transformers for language understanding, in *North American Chapter of the Association for Computational Linguistics (NAACL)* (Association for Computational Linguistics, 2019), pp. 4171–4186.
29. D. Yogish, T. N. Manjunath and R. S. Hegadi, Review on natural language processing trends and techniques using NLTK, in *Recent Trends in Image Processing and Pattern Recognition: Second Int. Conf., RTIP2R 2018*, Solapur, India, December 21–22, 2018, Revised Selected Papers, Part III 2 (Springer, Singapore, 2019), pp. 589–606.
30. V. Sanh, L. Debut, J. Chaumond and T. Wolf, DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter, preprint (2019), arXiv:1910.01108.
31. Y. Liu *et al.*, Roberta: A robustly optimized bert pretraining approach, preprint (2019), arXiv:1907.11692.
32. Z. Lan, M. Chen, S. Goodman, K. Gimpel, P. Sharma and R. Soricut, Albert: A lite bert for self-supervised learning of language representations, preprint (2019), arXiv:1909.11942.
33. A. Radford, J. Wu, R. Child, D. Luan, D. Amodei and I. Sutskever, Language models are unsupervised multitask learners, *OpenAI Blog* **1**(8) (2019) 9.
34. Y. Sun, S. Wang, Y. Li, S. Feng, H. Tian, H. Wu and H. Wang, Ernie 2.0: A continual pre-training framework for language understanding, in *Proc. AAAI Conf. Artificial Intelligence*, Vol. 34, No. 05 (AAAI, 2020), pp. 8968–8975.
35. K. Clark, Electra: Pre-training text encoders as discriminators rather than generators, preprint (2020), arXiv:2003.10555.
36. Z. Yang, XLNet: Generalized autoregressive pretraining for language understanding, preprint (2019), arXiv:1906.08237.
37. A. Sbei, K. ElBedoui, W. Barhoumi and C. Maktouf, Adaptive feature selection in PET scans based on shared information and multi-label learning, *Vis. Comput.* **38** (2022) 257–277.